

Baidu releases quantum machine learning toolkit on GitHub

Baidu Inc. has released a toolkit for quantum machine learning, called Paddle Quantum, on GitHub. The toolkit enables developers to build and train quantum neural network models. It has been built on Baidu's deep learning platform, PaddlePaddle.

Paddle Quantum includes a chemistry library with optimisation tools and three quantum apps — for machine learning, chemical simulation and combinatorial optimisation. Sources at Baidu said that Paddle Quantum can be used to support quantum circuit models and general quantum computing research.



Duan Runyao, director of Baidu's Institute for Quantum Computing, said that researchers in the quantum field can use Paddle Quantum to develop quantum artificial intelligence. He added that deep learning enthusiasts now have a shortcut to learning quantum computing.

Baidu also announced a set of new tools that will provide a total of 27 enhanced features for the PaddlePaddle framework. These include Paddle.js, which is a deep learning JavaScript library that enables developers to use AI within a browser or mini programs for applications like WeChat. It will include Parakeet, a text-to-speech toolkit that supports models like ClariNet and WaveFlow; as well as Paddle X, which is an integration development tool that helps in faster data processing.

The 'Veracode State of Software Security (SOSS): Open Source Edition' analysed the component open source libraries across the Veracode platform database of 85,000 applications. This accounted for 351,000 unique external libraries. It said that nearly all modern applications, including those sold commercially, are built using some open source components. It added that a single flaw in one library can cascade to all applications that leverage that code.

Chris Eng, chief research officer at Veracode said, "Open source software has a surprising variety of flaws. An application's attack surface is not limited to its own code and the code of explicitly included libraries, because those libraries have their own dependencies. In reality, developers are introducing much more code, but if they are aware and apply fixes appropriately, they can reduce risk exposure."

The research also pointed out that some language ecosystems tend to pull in many more transitive dependencies than others. It added that in more than 80 per cent of JavaScript, Ruby and PHP applications, the majority of libraries are transitive dependencies.

As per the research, the language selection makes a difference both in terms of the size of the ecosystem and in the prevalence of flaws in those ecosystems. Typically, any PHP library has over a 50 per cent chance of bringing a security flaw along with it.

The report stated, "Among the OWASP (Open Web Application Security Project) Top Ten flaws, weaknesses around access control are the most common, representing over 25 per cent of all flaws. Cross-site scripting is the most common vulnerability category found in open source libraries – found in 30 per cent of libraries – followed by insecure deserialisation (23.5 per cent) and broken access control (20.3 per cent)."

OSS will drive telecom sector's innovation agenda by 2025

Open source software (OSS) holds the potential to play a key role in telco cloud deployments, a market that will potentially grow to US\$ 29 billion by 2025, a report by ABI Research claims. It said that OSS is gradually driving the innovation agenda for communications service providers (CSPs). This is expected to pose a challenge to the dominance of proprietary solutions in the telecom industry.

The report said that OSS and, by extension, cloud technologies promise nimbleness, but whether CSPs can seize the opportunity is yet to be seen.

Don Alusha, senior analyst at ABI Research said, "Though CSPs are at different time slots in their digitalisation journey, they should collectively propel the open source agenda forward. A close collaboration between standards bodies and open source communities is a step in that direction."

The report said that in the telecom sector, the adoption of OSS is already underway among CSPs and it will almost certainly become mainstream by 2025. It said that CSPs no longer have reservations in adopting OSS but are now considering ways to include it in their network operations and commercial undertakings.

SUSE teams up with Elektrobit to transform how cars will operate

SUSE has announced a strategic partnership with Elektrobit to deliver a next-generation Linux based operating system for future intelligent vehicles. Elektrobit is an international supplier of embedded software solutions and services for the automotive industry.